Briefings in Bioinformatics, 2025, **26(4)**, bbaf324
https://doi.org/10.1093/bib/bbaf324
Problem Solving Protocol

AFToolkit: a framework for molecular modeling of proteins with AlphaFold-derived representations

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Abstract

A key challenge in protein engineering is understanding how mutations affect protein fitness and stability. Most of current state-of-the-art models fine-tune protein structure prediction or protein language models or even pretrain their own. Despite its widespread use within computational workflows, AlphaFold2 exhibits limited sensitivity in assessing the effects of amino acid point mutations on protein structure, thereby constraining its utility in sequence design and protein engineering. In this work, we propose a simple modification of AlphaFold2 inference that improves the model's capacity to capture the structural impacts of amino acid mutations. We achieve this by discarding the multiple sequence alignment and masking the template in recycling stages. Moreover, we introduce AFToolkit, a framework that leverages the embeddings of the modified AlphaFold2 model and simple adapter models to solve multiple protein engineering tasks. In contrast to other methods, our approach does not require fine-tuning the AlphaFold2 model or pretraining a new model from scratch on large datasets. It also supports handling multiple mutations, insertions, and deletions by directly modifying the input protein sequence. The proposed approach achieves strong performance across established benchmarks in terms of Spearman correlation: 0.68 on PTMul, 0.60 on cDNA-indel, and 0.57 on C380.

Keywords: protein engineering; protein stability; protein–protein binding affinity

Introduction

Protein engineering is crucial in biotechnology, medicine, and synthetic biology [1]. It involves modifying existing proteins or designing new ones to achieve specific properties [2, 3]. A common approach is targeted mutations, which impact stability, binding affinity, and functionality [4]. Thus, understanding how mutations affect the protein is crucial. Traditional experimental methods for studying protein mutations are often time-consuming and costly as they require wet lab experiments [5]. Despite recent advances in automating and accelerating this process [6–8], it is necessary to develop efficient computational approaches.

Deep learning has significantly advanced computational biology by providing protein modeling tools, such as AlphaFold2 [9], AlphaFold3 [10], and open-source implementations of these models [11–13]. Deep learning models have been successfully applied to tasks such as protein sequence recovery [14], predicting stability changes due to mutations [15–23], and assessing alterations in protein–protein interactions [24–29]. However, approaches that rely on fine-tuning existing protein structure or language models [18, 22, 28, 29] or on training entirely new models from scratch using large datasets [21, 23, 26, 30, 31] can be prohibitively expensive. Not only do these methods drive up costs, but they often yield lower performance in structure prediction than trusted baselines like AlphaFold2 and ESMFold [32] (see fig. 2 in [31]). Other approaches generate vast amounts of synthetic data [21, 33, 34] to address the scarcity of experimental data. This strategy also comes at a high computational cost and may ultimately compromise model quality due to synthetic data quality. Finally, some of the approaches [18, 24, 26, 30] are limited to single point mutations and cannot process multiple mutations, and most approaches are incapable of processing insertions and deletions.

Several approaches utilize embeddings of the AlphaFold2 model for mutation-related tasks [22, 28]. However, its standard implementation relies heavily on multiple sequence alignment (MSA) and template features during the recycling phase [35]. We hypothesize that this dependence on coevolutionary and template features is one of the reasons for AlphaFold2's limited predictive power in evaluating the effects of amino acid point mutations [36], possibly due to the predicted structures for mutants being highly correlated with wildtype structures and therefore not accounting for the effects of mutations. The AF2Rank [35] method involves excluding MSA precalculation and replacing template features with special GAP tokens, significantly improving the performance of AlphaFold2 in the decoy structure quality assessment task. Thus, we hypothesize that AF2Rank or a similar modification can also improve performance in mutation effect prediction tasks.

Received: December 5, 2024. **Revised:** May 12, 2025. **Accepted:** June 9, 2025

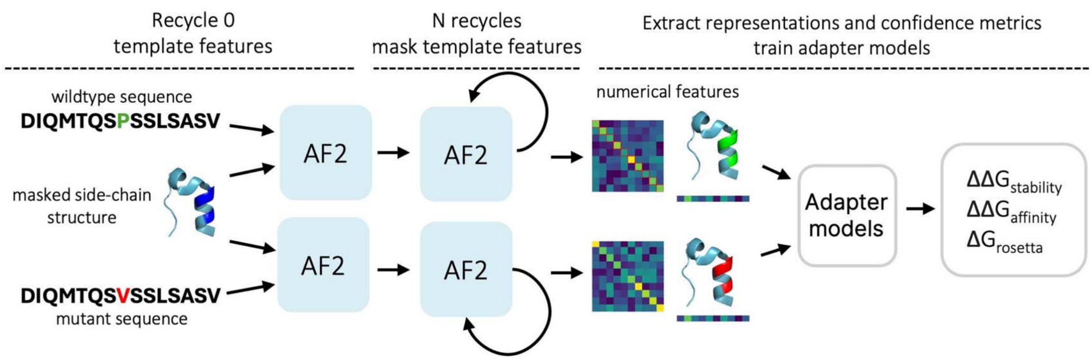

Figure 1. Scheme of the AFToolkit framework.

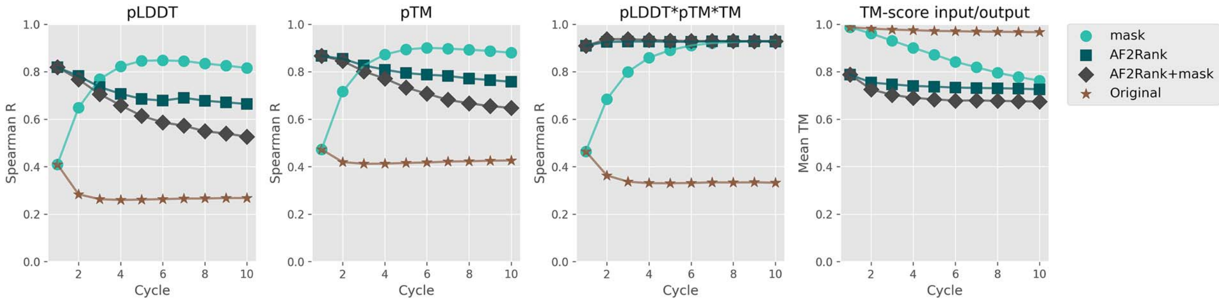

Figure 2. Evaluation of the proposed "mask" scheme on decoy ranking task for the randomly selected subset of 1800 samples of ROSETTA decoy dataset [44] with confidence metrics derived as described in [35] and TM score, representing the similarity between the input and output structures after model inference, provided on the right.

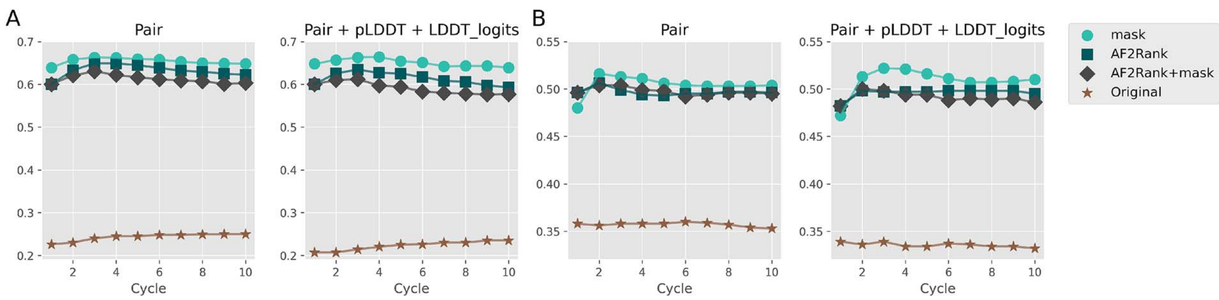

Figure 3. Spearman correlation between SVM models predictions and experimental values on (A) PTMul test set; (B) s669 test set, for models built upon embeddings extracted from AF2 using four approaches: vanilla AlphaFold2 without MSA, Mask, AF2Rank, AF2Rank + Mask and different AlphaFold recycling number from 1 to 10.

In this work, we investigate how modifying the template features affects the AlphaFold2's ability to predict effects of mutations by leveraging its internal embeddings. We propose a similar approach to AF2Rank that incorporates the input structure as a template, excludes MSA precalculation during model inference, and applies masking to the template features after the first AlphaFold inference cycle. We call this approach "mask." This approach is designed to preserve the input structural fold of the protein, which is an important quality in protein engineering tasks [14]. Importantly, our proposed method only modifies the inference process of AlphaFold2, requiring no retraining of the model. We show that such masking provides optimized confidence metrics (see Fig. 2) and better representations for downstream tasks, such as predicting changes in free energy ($\Delta\Delta G$) upon mutation (see Fig. 3). Moreover, the practical potential of the proposed masking is demonstrated in [37], where a very similar approach has been applied to design a ctenophore Ca^{2+}-regulated photoprotein berovin.

In this work, we propose the AFToolkit framework that benefits from improved representations of mutated proteins. We extract the embeddings from our modified AlphaFold2 and combine them with simple adapter models such as support vector machines (SVM), gradient boosting trees (GBT), and multilayer perceptron (MLP). These adapter models address mutation effect assessment tasks: prediction of protein stability change $\Delta\Delta G_{stability}$, and protein–protein binding affinity change $\Delta\Delta G_{affinity}$ due to mutations. AlphaFold2 embeddings allow us to naturally handle multiple mutations, insertions, and deletions by altering the input protein sequence. The simplicity of adapter models ensures computational efficiency and ease of implementation. Results of our computational experiments demonstrate that AFToolkit achieves performance comparable with or better than state-of-the-art (SOTA) specialized models across different datasets.

Additionally, we demonstrate that extracted embeddings can be utilized to predict per-residue Rosetta energies $\Delta G_{rosetta}$ [38]. Despite the fact that Rosetta energy function is an approximation of free energy that relies on numerous assumptions and exhibits limitations in accurately assessing the effects of mutations, it is used as a part of an ensemble approach or as an initial approximation before applying more precise yet computationally intensive methods [16]. The integration of Rosetta energy prediction model into the AFToolkit provides researchers with an additional proxy metric alongside other properties derived from AlphaFold2 in a single inference step. This removes the need to run multiple software tools, simplifying the workflows. It also demonstrates another practical use for AlphaFold2 beyond structure prediction, highlighting its broader potential for computational modeling.

Material and methods

Improved protein representations

Our first contribution is a modification to AlphaFold2's inference that improves structure quality assessment and allows us to better evaluate the effects of mutations on protein representations.

We exclude the time-consuming MSA precalculation step, so the MSA features are calculated solely based on the input sequence as in [35]. We use the wildtype protein structure as a template in the initial recycling round. In consequent recycling rounds, all the template features are multiplied by zero to reduce the effect of the wildtype structure on the resulting structure. We call this procedure "masking." Masking helps prevent the leakage of input structural information during AlphaFold's structural prediction, which is expected to result in more reliable estimations of the effects of point mutations.

For point mutations, all side-chain atoms of mutated residues are excluded from the template structure and the residue is renamed according to the mutation. For deletions, we exclude the residue from the protein and renumber subsequent residues. For insertions, we insert a "dummy" residue with masked coordinates and renumber subsequent residues. The input protein structure can be obtained from the Protein Data Bank (PDB) [39] or estimated with a protein structure prediction model such as AlphaFold2. For multichain proteins, we follow the approach from [11] and provide the complex as a single-chain input and adjust the positional encodings to introduce a gap of 25 residues between chains.

AFToolkit architecture

AFToolkit is a set of tools designed to analyze the effects of mutations on monomers and protein complexes. The illustration of the AFToolkit framework is provided in Fig. 1. It leverages an open-source implementation of the AlphaFold2 model (namely, the OpenFold model [13]) and Model 2 pTM model weights [9]. We chose the pTM version of Model 2 to facilitate direct comparisons with the confidence metrics used by [35] (see Fig. 2). Moreover, an experimental study [37] used confidence metrics to design a ctenophore Ca^{2+}-regulated photoprotein berovin. We elected to use Model 2 because it was trained with fewer MSAs, reducing its reliance on MSAs. Since our protocol does not use MSAs for mutant structure prediction, we found that Model 2 aligns more closely with our methodological setup. It then utilizes the modified inference procedure described in Section Improved protein representations to get protein representations and confidence metrics. Finally, the protein representations are utilized in adapter models to predict the change in protein stability and protein–protein complex affinity due to mutations ($\Delta\Delta$G), as well as per-residue Rosetta energy. The adapter models include SVM [40, 41], gradient boosting [42], and MLP. Below, we describe the data processing pipeline, training scheme, and adapter models.

AFToolkit sequence aggregation

Let us define the AlphaFold2 embedding for the protein sequence S of length L as $\mathbf{x} \in \mathbb{R}^{L\times d}$, where d denotes the size of per-residue embedding. Let **Mut** define the set of mutations. The mutations are defined by mutcodes—strings with the following format: $R_{\text{before}}\,\mathbf{POS}\,R_{\text{after}}$, where $R_{\text{before/after}}$ is the residue type before and after mutation, and **POS** is the position of mutation. Note that R_{after} can be "−" in case of a deletion, and consists of two amino-acid types in case of an insertion. Additionally, let S^{WT} and S^{MT} define the wildtype and mutated proteins, and $\mathbf{x}^{WT}$ and $\mathbf{x}^{MT}$ their features.

In our work, we consider several feature-aggregations methods: (1) average (or sum) of per-residue embeddings over the whole protein, and (2) average (or sum) of per-residue embeddings over the mutated residues. The per-residue embeddings are obtained by concatenating pair representations, LDDT logits, and pLDDT values predicted by AlphaFold2. We then concatenate the aggregated features for wildtype and mutated proteins and pass the resulting vector to adapter models (see Sections AFToolkit SVM and AFToolkit CatBoost, AFToolkit MLP). To simplify the notation, we consider the wildtype and mutated proteins to be of the same length L. However, the lengths of proteins can differ in case deletions or insertions are present in **Mut**. In case of global aggregation, we calculate the feature-vector as

$$\begin{aligned} &\mathbf{f} = \left(\frac{1}{L}\sum_{i=1}^{L}\mathbf{x}_i^{WT}, \frac{1}{L}\sum_{i=1}^{L}\mathbf{x}_i^{MT}\right) \\ &\quad \text{– Global Mean Aggregation (Mean-GA);} \\ &\mathbf{f} = \left(\sum_{i=1}^{L}\mathbf{x}_i^{WT}, \sum_{i=1}^{L}\mathbf{x}_i^{MT}\right) \\ &\quad \text{– Global Sum Aggregation (Sum-GA).} \end{aligned} \tag{1}$$

In case of mutation-based aggregation, for point mutations we get

$$\begin{aligned} &\mathbf{f} = \left(\frac{1}{|\mathbf{Mut}|}\sum_{i\in\mathbf{Mut}}\mathbf{k}_i, \frac{1}{|\mathbf{Mut}|}\sum_{i\in\mathbf{Mut}}\mathbf{k}_i\right) \\ &\quad \text{– Mutation Mean Aggregation (Mean-SA);} \\ &\mathbf{f} = \left(\sum_{i\in\mathbf{Mut}}\mathbf{k}_i, \sum_{i\in\mathbf{Mut}}\mathbf{k}_i\right) \\ &\quad \text{– Mutation Sum Aggregation (Sum-SA),} \end{aligned} \tag{2}$$

where $\mathbf{k}_i$ is the representation of position i. The $\mathbf{k}_i$ is calculated differently for various types of mutations:

$$\begin{aligned} &\mathbf{k}_i = \mathbf{x}_i \text{– point mutations;} \\ &\mathbf{k}_i = \left(\frac{1}{2}\left(\mathbf{x}_{i-1}^{WT} + \mathbf{x}_{i+1}^{WT}\right), \frac{1}{2}\left(\mathbf{x}_{i-1}^{MT} + \mathbf{x}_{i}^{MT}\right)\right) \\ &\quad \text{– deletions, } i \text{ is the index of deleted residue;} \\ &\mathbf{k}_i = \left(\frac{1}{2}\left(\mathbf{x}_{i}^{WT} + \mathbf{x}_{i+1}^{WT}\right), \frac{1}{2}\left(\mathbf{x}_{i}^{MT} + \mathbf{x}_{i+2}^{MT}\right)\right) \\ &\quad \text{– insertions, } i \text{ is the positions of insertion.} \end{aligned} \tag{3}$$

Additionally, for the mutation-based aggregation, we consider two training setups. The first setup includes all types of mutations in the training dataset. In inference, the features for multiple mutations are calculated according to Equation 2. In the second setup, we exclude multiple mutations from the training dataset. During inference, to predict the effect of multiple simultaneous mutations, we first process the mutated protein (with all mutations) with an AlphaFold2 model to get the embeddings of residues. We then calculate the effect of multiple mutations as the sum of the effects of single mutations. See example of prediction for protein with two mutations: **Mut** = {A15D, Y23T} in Equation 4.

$$\Delta\Delta\tilde{G}(S, S_{A15D,Y23T}) = \Delta\Delta\tilde{G}(S, S_{A15D}) + \Delta\Delta\tilde{G}(S, S_{Y23T}), \quad (4)$$

where $S_{A15D,Y23T}$ denotes the protein S with two mutations at positions 15 and 23, and $\Delta\Delta\tilde{G}$ is the predicted $\Delta\Delta G$. This approach allows us to use the inductive bias about the mostly additive nature of the effects of multiple point mutations while accounting for the epistasis effect: embeddings of mutated residues S_{A15D} and S_{Y23T} have exchanged information multiple times during inference of the AlphaFold2 and account for the joint effect of multiple mutations. We dub this approach as Single-MA.

AFToolkit SVM

We utilize the `scikit-learn` [43] implementation of Support Vector Regressor in all our computational experiments. We found this model to be robust and to exhibit strong performance across all datasets and tasks without the need for major retuning of the hyperparameters. Specifically, when applying the SVM to the protein–protein affinity change prediction, we only adjusted the regularization hyperparameter C ($C = 1$ for protein stability change prediction and $C = 2$ for protein–protein binding affinity change prediction). The SVM Regressor is used in combination with standard scaling data preprocessing.

For tasks of protein stability change prediction and protein–protein binding affinity change prediction, we apply the same approach of concatenating Pair representations, LDDT logits, and pLDDT values of mutated positions and using the resulting representations as SVM model input. Note that we do not use representations of individual chains of the protein–protein complex as input (see Section Improved protein representations).

AFToolkit CatBoost

CatBoost [45] implementation of gradient boosting on decision trees is used to train a regression model on the AFToolkit embeddings. To extract features from input proteins, we follow the setup described in Section AFToolkit SVM. The CatBoost model was trained to minimize mean squared error (MSE) for a maximum of 500 iterations with early stopping after 15 consecutive iterations without loss improvement. Other hyperparameters that differ from default values include a maximum depth or weak learners of 4, and L2 regularization coefficient of 5.

AFToolkit MLP

We consider the `scikit-learn` [43] implementation of the MLP as an AFToolkit adapter. The input features are extracted in the same way as for SVM and CatBoost models. We use MLP jointly with the Standard Scaler data preprocessing. We use default hyperparameters, except for the early stopping flag. The total number of training epochs, optimizer, batch size, and learning rate annealing technique. In our setup we train MLP for 50 epochs with a stochastic gradient descent (SGD) optimizer and batch size of 256. Learning rate is annealed exponentially from 10^{-3} and early stopping criteria is used. Our computational experiments demonstrated that training with Adam optimizer leads to overfitting, as can be seen in Supplementary table S4. We used the SGD optimizer instead.

Rosetta energy prediction

As opposed to predicting protein folding Gibbs free energy change due to mutation $\Delta\Delta G = \Delta G(S^{MT}) - \Delta G(S^{WT})$, as in previously described tasks, in this task we aim to predict $\Delta G(S)$ directly. We do so by treating all input protein residues as independent samples and using per-residue Rosetta energy function values as targets (see Section Datasets for details). For each residue, concatenation of Pair representations, LDDT logits, and pLDDT value (same representations as in Section AFToolkit sequence aggregation) is used as input to the SVM model. Note that this task performs per-residue prediction, and thus requires no protein-level embeddings aggregation.

Datasets

Stability change prediction

We selected a recent large experimental dataset cDNA [7] for training. We used the homology based [9, 46–48] train/test splits from [22]. Note that Mutate Everything [22] can only process point mutations, so insertions and deletions were excluded from their dataset. As our proposed model is capable of processing all types of mutations, we keep the PDBid-based splits from [22] while adding insertions and deletions for corresponding proteins. The resulting cDNA-test test dataset contains 50 005 samples. To thoroughly evaluate our model on various types of mutations, we split cDNA-test into various categories containing certain types of mutations: cDNA-1 for single point mutations, cDNA-2 for double point mutations, and cDNA-indel for insertions and deletions. Additionally, we select *de novo* proteins that were not included in the Mutate Everything dataset [22], filter them to exclude any homologies to the proteins in the training data, and group them into a new test dataset that we call *de novo* Mega. To demonstrate the robustness of our model on different types of proteins, we also employ widely used s669 [49] and ssym [50] as test sets. To test the performance of our models on multiple point mutations (up to 10 simultaneous mutations), we use the PTMul dataset.

Although extensive, the cDNA dataset contains only short proteins. To mitigate this issue, we extend the training set with proteins from the PROSTATA dataset [18]. We called our this extended dataset cDNA+PROSTATA. Following [22], we excluded proteins homologous to proteins from test datasets from the cDNA+PROSTATA dataset using BLAST [51], filtering out proteins with >36% sequence identity and an *E*-value < 0.05. The final train set contains 223 611 samples, including 221 236 samples from cDNA dataset and 2375 from PROSTATA dataset.

Due to the fact that the Gibbs free energy change due to mutation $\Delta\Delta G(S^{WT}, S^{MT})$ is an anticommutative function, i.e. $\Delta\Delta G(S^{WT}, S^{MT}) = -\Delta\Delta G(S^{MT}, S^{WT})$, we extend the cDNA+PROSTATA training dataset with reverse mutations, as previously done in [50, 52–54]. For example, if mutation *A15D* has a $\Delta\Delta G = -0.57$, than we add a new mutation *D15A* with $\Delta\Delta G = 0.57$. This allows us to additionally regularize the models.

Protein–protein interaction energy change prediction

To evaluate the performance of AFToolkit on protein complexes, we employ the M1101 (AB-bind) dataset [55] and S1131 and S4169 subsets from SKEMPI2 dataset [56]. An established way to evaluate models on these datasets is through cross-validation. We adopt the cross-validation splits from [28]. Moreover, it has recently been shown that mutation-level splits are prone to target leakage [29, 57]. Therefore, we additionally perform cross-validation on complex-based splits also taken from [28]. We also use the C380 dataset [58], which only contains SARS-COV2 S-protein and antibody complexes as an independent test set. This dataset was designed to test the ability of protein–protein binding affinity change models to generalize beyond the widely used SKEMPI-2 dataset. We train models on the S4169 dataset before evaluation on C380.

Rosetta energy prediction

Rosetta energy was calculated for a randomly selected set of 10 000 relaxed AlphaFold protein structures obtained from the AlphaFold database of UniProt structures [59]. Each structure was relaxed, by conducting 100 iteration of FastRelax module implemented in PyRosetta with constraints applied to native coordinates. REF2015 energy function was calculated for each residue in all proteins [16].

Metrics

To evaluate our models quantitatively, we follow [22] and use the following five metrics for all tasks. Spearman correlation coefficient R_s, Pearson correlation coefficient R_p, and RMSE between the $\Delta\Delta\tilde{G}$ and $\Delta\Delta G$ are used to evaluate the quality on regression tasks. Protein stability and binding affinity change prediction are sometimes viewed as classification tasks by considering mutations with $\Delta\Delta G < -0.5$ as stabilizing and mutations with $\Delta\Delta G \geq -0.5$ as destabilizing (the threshold is taken from [22]). For this reformulation, ROC-AUC (labeled AUC in tables) and MCC metrics are used.

Results

Improved protein representations

To demonstrate the performance of our proposed approach, we assess it in two tasks: (1) evaluating the correlation between AlphaFold2 confidence metrics and the similarity of decoys to the native structure, and (2) evaluating the quality of mutation effect assessment. We use another inference modification technique proposed in [35] called AF2Rank as a baseline. In this approach, the template tokens provided for the model inference are replaced by "gap" tokens.

In task (1), we reproduce the computational experiments described in the AF2Rank paper [35]. Specifically, we evaluate the ability of the proposed modified AlphaFold2 model to rank decoys $\mathbb{d}_{in}$ generated by the Rosetta software by their similarity to the experimental ground truth structures $\mathbb{e}$, as measured by the Template Modeling (TM) scores between them $TM(\mathbb{d}_{in}, \mathbb{e})$. This is done by taking the decoy structure $\mathbb{d}_{in}$ as template and considering the predicted Local Distance Difference Test (pLDDT) scores $pLDDT(\mathbb{d}_{in})$, predicted Template Modeling (pTM) scores $pTM(\mathbb{d}_{in})$, and TM scores $TM(\mathbb{d}_{in}, \mathbb{d}_{out})$ (which we call TM-score input/output in Fig. 2) between the input decoy structure $\mathbb{d}_{in}$ and AF2's output structure $\mathbb{d}_{out}$. To quantify this ranking propensity, we compute the Spearman's rank correlation coefficient (Spearman's R) between these scores and ground truth TM scores $TM(\mathbb{d}_{in}, \mathbb{e})$. The model demonstrates a stronger correlation between the confidence metrics, pLDDT and pTM scores, and the quality of the input structure. Overall, the "mask" scheme proposed in this study helps to better preserve the structural integrity of the analyzed input (see Fig. 2), as shown by comparing the TM score before and after model inference.

In task (2), we compare the embeddings from the vanilla AlphaFold2 model with no MSA, AF2Rank, and our proposed "mask" approach in the task of stability change prediction due to mutations. To train the models, we used the same hyperparameters for the SVM adapter model and only changed the input embeddings. Additionally, we benchmark the combination of the aforementioned methods that we call "AF2Rank+mask."

The results in Fig. 3 confirm the findings of the work by Pak *et al.* [36] and show that the embeddings of the vanilla AlphaFold2 model without an MSA do not always yield strong predictive power for amino acid stability changes. Furthermore, Fig. 3 and Supplementary material (Tables S1 and S2) show that the proposed "mask" approach provides embeddings that lead to better metrics on a downstream $\Delta\Delta G$ change prediction task than the AF2Rank method.

Based on the results in Fig. 2 and Supplementary material (Tables S1 and S2), we choose the number of recycles $\mathbf{N} = 4$ and the features used by all adapter models to be the concatenation of Pair, pLDDT, and LDDT logits. At this number of recycles, the model provides comparable performance in the decoy ranking task relative to the AF2Rank method while having a more negligible effect on the input structure. This choice is motivated by the need to preserve the user-defined conformational state of the protein while allowing for an evaluation of the impact of mutations. It is particularly useful for the iterative optimization of proteins with a desired fold. For example, it facilitates the optimization of proteins with specific catalytic site geometries by guiding subsequent rounds of directed evolution. This is achieved by analyzing the output of AlphaFold inference, including amino acid residue geometries, confidence metrics, and $\Delta\Delta G$ features. Notably, we recently applied this approach to optimize the physico-chemical properties of the ctenophore Ca^{2+}-regulated photoprotein berovin in the apo conformational state, which resulted in successful experimental validation [37].

Ablation study

We begin with a comprehensive ablation study of various feature-aggregation methods, training configurations, and adapters. All models are trained on the full-scale dataset described in the Improved protein representations section and evaluated on the designated test sets. Figure 4 shows Spearman correlation results, while additional metrics are provided in the Supplementary Material (Table S3). Guided by these findings, we adopt the Single-MA approach alongside an SVM adapter as our primary model, denoted AFToolkit (SVM).

Comparison of sequence-based and structure-based approaches

To justify the use of AlphaFold2 embeddings, which are reasonably compute-expensive to obtain, we first compare the performance of AFToolkit models with sequence-based and structure-based feature encoders on the cDNA+PROSTATA dataset. The ESM-2 [32] model is used as a sequence-based encoder. Note that we do not consider ESMFold a purely sequence-based model, as it has been trained on structures from the PDB, which contain many of the proteins from our test datasets. Moreover, our improved AlphaFold2 inference procedure that discards the MSA nearly matches ESMFold's computational complexity. The embeddings obtained by different encoders are then used with the same SVM

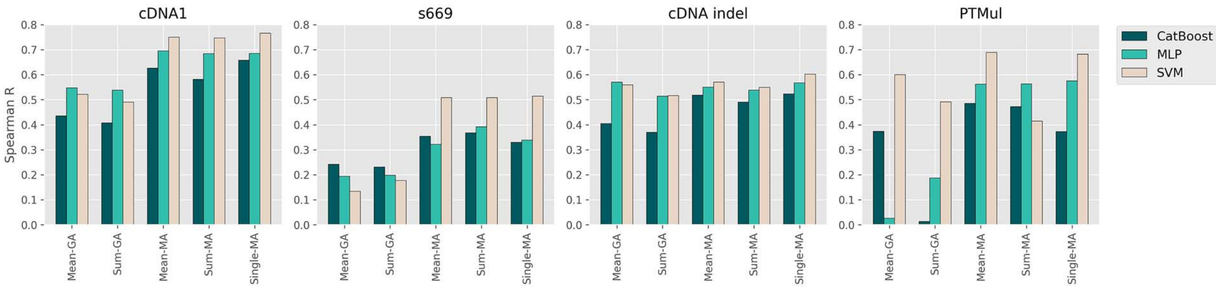

Figure 4. Adapter models trained on cDNA+PROSTATA dataset for monomer mutation stability change prediction on cDNA1, s669, cDNA-indel, and PTmul test sets using four different approaches for embedding aggregation and the full training dataset (Global Mean Aggregation (Mean-GA), Global Sum Aggregation (Sum-GA), Mutation Mean Aggregation (Mean-MA), and Mutation Sum Aggregation (Sum-MA)), as well as the cDNA+PROSTATA dataset without multiple mutations (Single-MA).

Table 1. Comparison of structure and sequence-based models

Method	R_s	R_p	**AUC**	**MCC**	**RMSE↓**
cDNA-1 (single mutations only)					
SVM (AF2)	**0.77**	**0.77**	0.83	0.25	**0.77**
SVM (ESM2)	0.71	0.70	0.81	0.20	0.92
ME (ESM2)	0.69	0.71	0.76	0.19	0.86
PROSTATA	0.76	0.75	**0.86**	**0.34**	0.85
cDNA-2 (double mutations only)					
SVM (AF2)	**0.57**	**0.59**	**0.86**	**0.22**	1.22
SVM (ESM2)	0.44	0.44	0.75	0.05	**1.16**
ME (ESM2)	0.36	0.39	0.75	0.09	1.54
PROSTATA	0.46	0.47	0.77	0.06	1.25
s669					
SVM (AF2)	**0.51**	**0.52**	**0.74**	**0.20**	**1.41**
SVM (ESM2)	0.44	0.41	0.70	0.16	1.56
ME (ESM2)	0.41	0.39	0.65	0.08	1.54
PROSTATA	0.47	0.46	0.66	0.14	1.50

adapter model. We dub the resulting models as SVM-AF2 and SVM-ESM2. See the results in Table 1.

Second, we compare the AFToolkit model that utilizes AlphaFold2 embeddings with other sequence-based baselines: PROSTATA [18] and Mutate Everything with an ESM-2 backbone (ME-ESM) [22]. Both the PROSTATA and Mutate Everything models were retrained from scratch on our cDNA+PROSTATA dataset. The reverse mutations were excluded from the dataset when training ME. The ME is designed to simultaneously predict the effects of multiple mutations on a single protein, which mimics the experimental setup of the cDNA dataset. However, when applying ME to reverse mutations from the cDNA dataset, the total amount of unique input proteins increases by ∼800 times, which makes the training of ME inefficient and extremely resource-intensive. See the comparison in Table 1.

Finally, we compare our proposed approach with sequence-based methods on protein–protein interaction energy change prediction. As baselines for this task, we use the AFToolkit with a ESM-2 feature encoder and the MuLAN model [29]. See the results in Table 2.

Protein stability change prediction

Following recent work on predicting the stability change due to mutations [18, 22], we compose a new dataset that consists of cDNA dataset [7] and PROSTATA dataset [18] and dub it cDNA+PROSTATA. The cDNA dataset contains single and double point mutations, insertions, and deletions, whereas the PROSTATA dataset contains large proteins that are not present in cDNA. We evaluate the AFToolkit models and several strong baselines [18, 22] on commonly used test sets [7, 49, 50, 60]. To ensure a fair comparison with peer reviewed methods, we retrain PROSTATA [18] and MutateEverything [22] on our proposed cDNA+PROSTATA dataset. The predicted $\Delta\Delta G$ for PROSTATA model on cDNA-2 and PTMul is calculated by summing the effects of single mutatations that compose the multiple mutation (see Equation 4).

The results show that AFToolkit with an SVM adapter is better than or comparable with baselines one on all datasets (Table 3 and Table 4). AFToolkit performs on par with SOTA methods on single mutation datasets (cDNA-1, ssym, s669, *de novo* Mega), and exceeds their performance on multiple mutation datasets (cDNA-2, PTMul). Moreover, it is the only approach that is capable of processing insertions and deletions.

We study in detail the ability of AFToolkit to predict effects of multiple mutations. We consider the protein lysozyme T4 (PDB ID:2LZM) samples from the PTmul test set: seven samples in total, corresponding to different combinations of seven point mutations. We predict the effect mutations in two approaches: (1) we score each point mutation independently and then sum the corresponding effects, and (2) as described in the Materials and Methods section, where we score a simultaneous effect of multiple point mutations. We observe that approach 2 shows a more precise effect ($RMSE_1$=1.12 versus $RMSE_2$=0.99, Table S5). For lysozyme T4, these mutations are additive according to the experimental data [61], yet our approach performs better when predicting multiple mutations than adding effects of single ones. We expect this to be the case, since AlphaFold's

Table 2. Comparison of the Pearson correlation (R_p) of the models on PPI ddG datasets using 5-fold protein complex-based cross-validation with metrics for models marked with $*$ extracted from paper [28]

Model	S1131	S4169	M1101
Complex-based splits			
SVM (AF2)	0.73 ± 0.04	**0.67** ± **0.05**	**0.47** ± **0.28**
SVM (ESM2)	0.55 ± 0.05	0.37 ± 0.21	0.07 ± 0.37
MuLAN-Ankh	**0.77** ± **0.06**	0.58 ± 0.14	0.30 ± 0.34
MuLAN-ESM	0.76 ± 0.07	0.60 ± 0.12	0.17 ± 0.27

Table 3. Comparison of monomer ΔΔG prediction models trained on cDNA+PROSTATA dataset using R_s - Spearman correlation and R_p - Pearson correlation coefficient

Model	R_s	R_p	AUC	MCC	RMSE↓
cDNA-1 (single mutations only)					
ME-AF2	**0.81**	**0.81**	0.82	0.21	**0.70**
ME-ESM2	0.69	0.71	0.76	0.19	0.86
PROSTATA	0.76	0.75	**0.86**	0.34	0.85
AFToolkit (SVM)	0.77	0.78	0.83	**0.38**	0.74
cDNA-2 (double mutations only)					
ME-AF2	0.52	0.55	0.79	0.18	**1.22**
ME-ESM2	0.36	0.39	0.75	0.09	1.54
PROSTATA	0.46	0.47	0.77	0.06	1.25
AFToolkit (SVM)	**0.57**	**0.59**	**0.86**	**0.22**	**1.22**
cDNA-indel (insertions and deletions only)					
AFToolkit (SVM)	**0.60**	**0.62**	**0.89**	−**0.002**	**1.02**
s669					
ME-AF2	**0.54**	**0.53**	**0.74**	0.17	**1.40**
ME-ESM2	0.41	0.39	0.65	0.08	1.54
PROSTATA	0.47	0.46	0.66	0.14	1.50
AFToolkit (SVM)	0.51	0.52	**0.74**	**0.20**	1.41
Ssym					
ME-AF2	0.61	0.65	0.76	0.30	1.23
ME-ESM2	0.36	0.39	0.58	−0.03	1.49
PROSTATA	0.50	0.54	0.72	0.15	1.42
AFToolkit (SVM)	**0.77**	**0.76**	**0.83**	**0.36**	**1.10**
PTMul					
ME-AF2	0.67	**0.68**	0.87	**0.53**	**1.71**
ME-ESM2	0.43	0.37	0.76	0.41	2.18
PROSTATA	0.45	0.57	0.80	0.44	2.12
AFToolkit (SVM)	**0.68**	0.60	**0.89**	0.52	1.90
de novo Mega					
ME-AF2	0.63	0.67	0.71	0.07	0.66
ME-ESM2	0.64	0.66	0.74	0.11	0.74
PROSTATA	**0.72**	**0.75**	**0.89**	**0.33**	0.62
AFToolkit (SVM)	0.71	0.73	0.86	0.28	**0.61**

structure prediction is less sensitive to single mutations than multiple.

Protein–protein interaction energy change prediction

To evaluate the performance of AFToolkit on protein complexes, we employ the M1101 (AB-bind) dataset [55] and S1131 and S4169 subsets from SKEMPI2 dataset [56]. We adopt both the mutation-level and complex-based cross validation splits from [28] to reduce the effect of target leakage [29, 57]. As baselines, we use models based on different approaches, such as graph-based MpbPPI [28], GeoPPI [25] and mCSM-PPI2 [24], topological TopNetTree [30] and PerSpect-EL [26], force field-based FoldX [15], and pLM-based MuLAN [29]. Table 5 shows that **AFToolkit** performs comparably with SOTA specialized models.

We also use C380 dataset [58] as an independent test set. Table 6 demonstrates that while our proposed AFToolkit model shows a performance drop compared with AB-bind and SKEMPI2 datasets, it remains a good predictor, on par with SOTA methods.

Rosetta energy prediction

The trained SVM model achieved a correlation coefficient of R=0.842±0.004 in predicting Rosetta Energy.

Discussion

In this work, we introduce an AFToolkit framework, which consists of a modified inference scheme for the pretrained AlphaFold2 model and several adapter models. Our approach does not require fine-tuning the AlphaFold2 model and can be extended to other

Table 4. Performance of the models (structure-based methods are in italic, the rest are sequence-based) on the S669 dataset with metrics for reviewed models taken from [18, 22, 49], and SPIRED* metrics differing from the original paper values due to being calculated per-protein and then averaged

	Direct		**Reverse**	
Model	R_s	**RMSE**	R_s	**RMSE**
AFToolkit (SVM)	0.51	1.41	0.51	**1.41**
ACDC-NN	0.46	1.49	0.45	1.5
DDGun3D	0.43	1.6	0.41	1.62
PremPS	0.41	1.5	0.42	1.49
ThermoNet	0.39	1.62	0.38	1.66
Rosetta	0.39	2.7	0.4	2.68
Dynamut	0.41	1.6	0.34	1.69
INPS3D	0.43	1.5	0.33	1.77
SDM	0.41	1.67	0.13	2.16
PoPMuSiC	0.41	1.51	0.24	2.09
MAESTRO	0.50	1.44	0.2	2.1
DUET	0.41	1.52	0.23	2.14
PROSTATA	0.49	1.45	0.49	1.45
Mutate Everything	**0.53**	**1.38**	0.49	1.48
INPS-Seq	0.43	1.52	0.43	1.53
ACDC-NN-Seq	0.42	1.53	0.42	1.53
DDGun	0.41	1.72	0.38	1.75
SPIRED*	**0.53**	1.48	**0.53**	1.48

Table 5. Comparison of the Pearson correlation (R_p) of the models on PPI ddG datasets using 10-fold mutation-level cross-validation and five-fold protein complex-based cross-validation, where metrics for ∗-marked models are extracted from paper [28], metrics for ∗∗-marked models are our approximations based on graphs from the same paper, as exact values of Pearson correlation for complex-based splitting are not reported in the paper; and †-marked predictions for mCSM-PPI2 are obtained from theofficialwebserver without retraining by summing independent single mutation effect predictions

Model	**S1131**	**S4169**	**M1101**
Mutation-based splits			
MpbPPI∗	0.87 ± 0.01	0.80 ± 0.01	0.79 ± 0.01
GeoPPI∗	0.85	0.78	0.78
TopNetTree∗	0.85	0.79	NA
PerSpect-EL∗	0.85	NA	NA
FoldX∗	0.42	0.33	0.27
mCSM-PPI2	0.80 ± 0.02	0.76 ± 0.04	0.57 ± 0.09 †
MuLAN-Ankh	0.83 ± 0.05	0.74 ± 0.03	0.64 ± 0.13
MuLAN-ESM	0.83 ± 0.05	0.73 ± 0.03	0.46 ± 0.18
AFToolkit (SVM)	0.82 ± 0.04	0.76 ± 0.04	0.77 ± 0.01
Complex-based splits			
MpbPPI∗∗	≈ 0.47	≈ 0.45	≈ 0.43
FoldX∗∗	≈0.42	≈ 0.33	≈ 0.27
mCSM-PPI2	0.80 ± 0.04	0.77 ± 0.04	0.54 ± 0.35 †
MuLAN-Ankh	0.77 ± 0.06	0.58 ± 0.14	0.30 ± 0.34
MuLAN-ESM	0.76 ± 0.07	0.60 ± 0.12	0.17 ± 0.27
AFToolkit (SVM)	0.73 ± 0.04	0.67 ± 0.05	0.47 ± 0.28

Table 6. Comparison of PPI mutation binding affinity change prediction models on the C380 dataset

Model	R_s	R_p	**AUC**	**MCC**	**RMSE↓**
FoldX	0.555	0.322	0.770	**0.330**	3.001
mCSM-PPI2	0.479	0.457	0.676	0.036	1.272
MuLAN-ESM	−0.005	−0.039	0.513	−0.015	1.452
MuLAN-Ankh	0.018	0.035	0.510	−0.015	1.463
AFToolkit (SVM)	**0.572**	**0.525**	**0.800**	0.239	**1.612**

mutation-related tasks by simply computing the embeddings for the dataset and training a light-weight adapter model on it. For example, the ProteinGym [62] benchmark contains tasks such as enzyme activity change prediction, expression change prediction, and binding specificity change prediction.

Our results of exploring protein representations confirm that the embeddings of the vanilla AlphaFold2 model without MSA input do not always yield strong predictive power for amino acid stability changes. We hypothesize that this limitation arises from the model's training process. Specifically, the heavy weighting of template features in the loss function may cause the model to become overconfident in its predictions whenever the output structure closely resembles the template. At the same time, our proposed "mask" approach facilitates the embeddings that lead to better metrics on $\Delta\Delta G$ prediction task than both vanilla AlphaFold and AF2Rank. We attribute this improvement to the impact of template features in each approach. Specifically, in AFToolkit, the number of recycles before masking can be adjusted to optimize the amount of structural information the model receives from the template, which prevents the model from becoming overconfident due to the high similarity of the output structure to the input template. In contrast, AF2Rank applies stronger masking to template input by replacing amino acid residues with UNK tokens, which cannot be adapted for improved performance in prediction of mutation effects.

Currently, we have only tested simple adapter models, such as SVM, GBT, and MLP. However, more sophisticated models that process embeddings of the entire sequences have the potential to significantly improve performance. The most obvious choice would be a transformer model [63]. Additionally, AFToolkit could be transformed into a specialized model by jointly fine-tuning the folding model alongside the adapter model. However, this approach is highly resource-intensive and prone to catastrophic forgetting, which may reduce the embeddings' applicability to other tasks.

Although versatile, performance of AFToolkit relies heavily on the quality of the structural input. Significantly inaccurate initial structural predictions may limit the tool's ability to effectively predict the effects of amino acid point mutations. However, high-accuracy tools, such as AlphaFold2 with default settings, can be used to generate reliable input structures. Looking ahead, next-generation models like AlphaFold3 [10] could further enhance structural accuracy and broaden the toolkit's applicability.

Overall, we expect the model to have broad applications in structure-guided protein sequence optimization, directed evolution algorithms, and as a complement to inverse folding models like ProteinMPNN [14]. Its key advantage over similar methods lies in its ability to preserve structural input while simultaneously predicting the effects of point mutations and providing opportunities to analyze the AFToolkit-refined mutated structure for subsequent rounds of optimization. Additionally, the predictions generated by multiple adaptor models, combined with model confidence metrics within a single run of the AFToolkit, offer valuable features for training structure-activity models. Thus, we anticipate that this model will become a highly effective tool for integration into protein engineering pipelines.

Conclusion

We proposed a modification to AlphaFold2 inference that allows improved protein representations. We then demonstrated that an SVM adapter model that utilizes these representations performs on par or better with SOTA specialized models on tasks such as predicting the stability change of monomers and protein–protein binding affinity due to mutations and per-residue Rosetta energy prediction. Unlike other models, our approach does not require a fine-tuning protein folding model and can handle multiple mutations, insertions, and deletions.

Key Points

- AFToolkit provides a modification of the AlphaFold2 inference pipeline that allows it to better capture the impact of single and multiple mutations.
- We evaluate our proposed modification against the original inference pipeline and the AF2Rank modification on Rosetta decoy ranking task and show improvement.
- We introduce a framework that utilizes protein embeddings from this modified pipeline and simple adapter models to solve protein engineering tasks (namely, prediction of protein stability change and protein–protein binding affinity change upon mutation) better than or on par with current SOTA methods.
- Unlike other models, our approach does not require fine-tuning a protein structure prediction model and can handle multiple mutations, insertions, and deletions.

Acknowledgments

The authors thank the anonymous reviewers for their valuable suggestions.

Author contributions

Maria Sindeeva (Data curation, Investigation, Software, Formal analysis, Visualisation, Writing—original draft), Alexander Telepov (Investigation, Formal analysis, Visualisation, Writing—original draft), Nikita Ivanisenko (Conceptualization, Formal analysis, Investigation, Writing—original draft), Tatiana Shashkova (Data curation, Investigation, Formal analysis, Software, Visualisation, Writing—original draft), Kuzma Khrabrov (Formal analysis, Writing—review & editing), Artem Tsypin (Formal analysis, Supervision, Writing—original draft, Writing—review & editing), Artur Kadurin (Supervision), Olga Kardymon (Supervision)

Supplementary data

Supplementary data is available at *Briefings in Bioinformatics* online.

Competing interests

No competing interest is declared.

Funding

This research did not receive external funding.

Data availability

The source code is available on GitHub at https://github.com/AIRI-Institute/AFToolkit.

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Briefings in Bioinformatics, 2025, **26(4)**, bbaf324
https://doi.org/10.1093/bib/bbaf324
Problem Solving Protocol